

Daniyal Khan

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Professional Summary:

AI Fullstack Engineer with hands-on experience designing and deploying **end-to-end computer vision and intelligent automation systems** that optimize workflows, enhance client engagement, and scale operations. Skilled in building **CV pipelines, LLM-powered systems, RAG architectures, AI copilots, and n8n workflow automations**. Proven track record of delivering **production-ready AI web platforms**, integrating **multi-agent systems with real business data**, and implementing **multi-tenant SaaS solutions**. Open to **remote and on-site opportunities**.

Core Skills:

AI / ML: LLMs, RAG Pipelines, Embeddings, Vector Search, Prompt Engineering, Agentic AI, Computer Vision, YOLO, FaceNet, Scikit-learn, Model Deployment

Programming : Python, C++, Pandas, NumPy, Scikit-learn, Machine Learning & Classification Models, LangChain, Ultralytics

Web & API Development: FastAPI, Flask, React, Tailwind, REST API Design

Integrations: Twilio (Voice/SMS Bots), Streamlit, External APIs (SERPAPI, Gemini API)

Databases & Data Management: MySQL, PostgreSQL, SQLite, Data Processing & ETL

Deployment & Tools: Docker, GitHub Actions, Render, Basic AWS/GCP

Version Control & CI/CD: Git, GitHub

Professional Experience:

2 years experienced Developer

AI Automation & Fullstack Engineer:

TechnoHub AI | 2024 – Present | Haripur / Remote (Hybrid)

- Designed, developed, and maintained **AI-driven automation systems** for startups and enterprise clients.
- Built **production-ready AI copilots and conversational systems** using **Gemini API, FastAPI**, and **REST-based architectures**.
- Developed **multi-tenant SaaS AI services** with **role-based access control** and deployed scalable solutions on **Render**.

- Integrated **workflow automation pipelines** using **n8n** and **Twilio**, enabling intelligent messaging, alerts, and business process automation.
- Collaborated with cross-functional teams to deliver AI solutions that **improved client response time and operational efficiency by up to 40%**.
- Contributed to system design, **API optimization**, and deployment strategies for **reliable, production-grade AI platforms**.

Freelance AI Automation Developer:

Mar 2023 – Mar 2024 | Remote

- Delivered **full-stack AI automation projects** for global clients across **e-commerce, CRM, and productivity platforms**.
- Designed and deployed **RAG-powered chatbots, workflow automations, and lead-generation systems** integrating **web scraping and email outreach pipelines**.
- Automated repetitive data handling and reporting tasks, **reducing manual client effort by up to 50%**.
- Developed **reusable Python modules** for **API orchestration, LLM integration, and real-time chatbot responses**, accelerating deployment across multiple clients.

Highlighted Projects:

Productix Copilot – AI-Powered Production Intelligence Platform:

Stack: FastAPI, PostgreSQL, React, Tailwind, Gemini API

- Developed a **multi-tenant SaaS platform** for analyzing **batch- and shift-level production data**.
- Designed a **three-level RBAC system** supporting subscriptions and enterprise account management.
- Built **Platform Manager workflows** to onboard companies, manage subscriptions, and control account activation/deactivation.
- Enabled **Company Manager roles** to create and manage multiple employee accounts.
- Developed an **AI-driven analytics engine** detecting productivity anomalies and generating actionable efficiency recommendations.
- Implemented a **Productivity Calculator** computing **single and combined productivity scores** with **AI recommendations**.
- Designed **AI agent workflows** for automated performance reports, improvement plans, and shift comparison analysis.
- Integrated a **RAG-based chatbot** using Gemini API to provide contextual production insights.
- Supported **Excel-based data ingestion** and automated report generation with download support.
- Delivered **batch-wise, shift-wise, and combined performance reports** through interactive dashboards.
- Built **REST APIs** with FastAPI and a responsive frontend with React and Tailwind.
- Designed scalable **PostgreSQL multi-tenant architecture** with secure tenant-level data isolation.

AI Trade Portal – GenAI-Powered CRM & Recommendation System:

Stack: Flask, Python, HTML, Gemini API, SERPAPI

- Built an **AI trade portal** recommending products and companies based on **user profiles, industry, and historical data**.
- Implemented a **RAG-based chatbot** using Gemini API to provide **context-aware trade insights** and reduce hallucinations.
- Designed **document ingestion and retrieval pipelines**, embedding trade data and company profiles for semantic search.
- Integrated **SERPAPI-based Google scraping** for up-to-date market and product information.
- Combined **CRM data with external sources** to generate **context-aware AI responses**.
- Developed **CRM workflows** for managing users, companies, leads, and interactions.
- Built REST APIs in Flask and supporting Python services for data processing and integrations.
- Ensured **modular architecture** for future scalability and model upgrades.

Face Recognition, Attendance & Visitor Management System:

- Built **real-time touchless attendance** where first appearance marks entry and last appearance marks exit.
- Implemented **YOLO (Ultralytics, PyTorch)** for person detection and **ResNet-SSD (Caffe)** for face localization.
- Used **FaceNet (InceptionResNetV1)** to generate **512-D embeddings** for identity recognition.
- Added **line-crossing/no-go area detection**, triggering alerts with captured face snapshots.
- Developed **visitor scheduling** with pre-registered face and CNIC photos.
- Integrated **gate camera detection** and **human verification loop** for guards to authorize visitors.
- Built **per-minute presence logging** and **automated stolen-time calculation** excluding breaks.
- Implemented **dynamic schedule and zone updates** for controlled access.
- **Containerized** the system using Docker and deployed with MySQL.
- Optimized **CPU-only inference** for cost-effective deployments.

Intelligent Document Search Application

- Built a **semantic search** engine over business documents using Python, **FAISS**, and **SentenceTransformers**, achieving fast retrieval of relevant content.
- Reduced average document query response time by 70% through **Redis caching** and optimized hybrid search.
- Integrated backend with frontend **React** interface, enabling interactive querying and visualization of results.
- Applied NLP techniques for document chunking and **embedding** to improve search accuracy and relevance.

Autonomous AI Workflow Agent:

- Engineered an end-to-end **AI agent** that converts natural language emails into structured **workflow actions**.
- Implemented Named **Entity Recognition** using **BERT-based transformer** model (HuggingFace).
- Built task classification logic for **automated** decision routing.
- Integrated **webhook**-based event-driven architecture using **n8n**.
- Automated Google Calendar event creation via **Oauth2**-secured **API integration**.
- Designed **modular agent** pipeline (Email → NLP → Decision → Execution).

Other Highlighted Projects:

- **Smart CRM System:** Full-stack CRM with **AI-driven lead scoring, complaint routing, and automated outreach** via n8n, improving sales efficiency.
- **Bike Rental Demand Predictor:** ML forecasting application predicting demand trends using **Scikit-learn**, deployed via **Streamlit** with interactive dashboards.

Key Achievements:

- Deployed **production-ready AI platforms** serving global clients across **e-commerce, CRM, and productivity domains**.
- Automated workflows, **reducing manual effort by 40–50%** and increasing operational efficiency.
- Maintained a **100% on-time delivery record** for freelance AI and automation projects.
- Recognized by **TechnoHub AI** for **innovative workflow automation solutions** enhancing client productivity.

Education:

Bachelor's in Artificial Intelligence | University of Haripur (2023 – Present)

Current CGPA: 3.75 / 4.00

Available for full-time work

Certifications:

- **Machine Learning** – Coursera
- **NumPy & Pandas** – Coursera
- **Scikit-learn** – Coursera
- **Agentic AI** – [DeepLearning.ai](https://www.deeplearning.ai)
- **IELTS** – Overall Band 7.5
- **Game Development:** NAVTEC